

Class Reg Number

Candidate Name _____

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MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Exam
Higher 2

CHEMISTRY

9647/03

Paper 3 Free Response

12 September 2011

2 hours

Additional Materials: Data Booklet
Writing paper

INSTRUCTIONS TO CANDIDATES

Write your name, class and register number in the spaces provided at the top of this page.

Answer 4 out of 5 questions in this paper.

Begin each question on a fresh page of writing paper.

Fasten the writing papers behind the given **Cover Page for Questions 1 & 2** and **Cover Page for Questions 3, 4 & 5** respectively.

Hand in Questions 1 & 2 and 3, 4 & 5 separately.

You are advised to spend about 30 min per question only.

INFORMATION FOR CANDIDATES

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You are reminded of the need for good English and clear presentation in your answers.

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- 1(a)** Lattice energies are often used as a measure of the energetic stability of ionic compounds. Usually ionic compounds with the same crystal structure can be compared based on the lattice energy value.

Lattice energies need to be determined experimentally using thermochemical data. However based on electrostatic considerations (i.e. attraction and repulsion of ions), theoretical values of the lattice energies can be calculated using equations such as the Kapustinskii equation:

$$\Delta H_{latt} = -107.1 \times \frac{\nu \times z_+ \times z_-}{r_+ + r_-} \text{ kJ mol}^{-1}$$

where ν is the number of ions in the empirical formula
 z_+, z_- is the charge of the cation and anion respectively
 r_+, r_- is the ionic radius (in nm) of the cation and anion respectively

The table below shows the numerical values of lattice energies for a series of magnesium and other related compounds.

Compound	LiCl	MgCl ₂	MgBr ₂	MgI ₂	MgO
Experimental value / kJ mol ⁻¹	- 853	- 2526	- 2440	- 2327	- 3933

- (i) Define, with the aid of an equation, the lattice energy of magnesium oxide.
- (ii) By quoting appropriate data from the *Data Booklet*, explain why the lattice energies of the magnesium halides decrease from MgCl₂ to MgI₂.
- (iii) By using appropriate data from the *Data Booklet*, calculate the theoretical lattice energy of magnesium chloride.
- (iv) The lithium and magnesium ion have similar ionic radii. Based on this assumption, use the value calculated in (iii) to estimate the theoretical lattice energy of lithium chloride.
- (v) Explain the differences between the theoretical and experimental values for lithium chloride and magnesium chloride respectively, and comment whether the assumption made in (iv) is a valid one.

[10]

- (b) Describe, and suggest an explanation for, the trend observed in the thermal decomposition of nitrates of the Group II elements from magnesium to barium. Write a general equation for the reaction that occurs on heating.

[3]

- (c) In an experiment, water is added to a test tube containing the solid remaining after magnesium carbonate has been heated. Dilute sulfuric acid is then added to the test tube. The procedure is repeated to the solid remaining after barium carbonate has been heated.

Describe and explain what you would see in both experiments.

[3]

- (d) The hexadentate ligand EDTA^{4-} forms a complex both with magnesium ions and calcium ions. With a suitable indicator, this can be used to determine the total concentration of magnesium ions and calcium ions in a mixture.

A sample of a mixture of anhydrous magnesium chloride and anhydrous calcium chloride had a mass of 0.344 g and was dissolved in water to make 100 cm^3 of solution. A 10 cm^3 portion of this solution required 33.3 cm^3 of $0.010 \text{ mol dm}^{-3} \text{ EDTA}^{4-}$ for complete reaction.

- (i) Suggest the formulae of both complexes formed.
- (ii) Calculate the mass of the anhydrous magnesium chloride in the mixture.

[4]

[Total: 20]

- 2(a)** A mixture was prepared using 1.00 mol of ethanedioic acid and 2.00 mol of ethanol. At a given temperature, the mixture was left to reach *dynamic equilibrium* according to the following equation.



The equilibrium mixture contained 36.8 g of ethanol.

- (i) What do you understand by the term *dynamic equilibrium*?
- (ii) Write an expression for K_c for this reaction.
- (iii) Use your expression in **(a)(ii)** to calculate the value of K_c .
- (iv) For this equilibrium, predict the effect of an increase in temperature on the value of K_c .

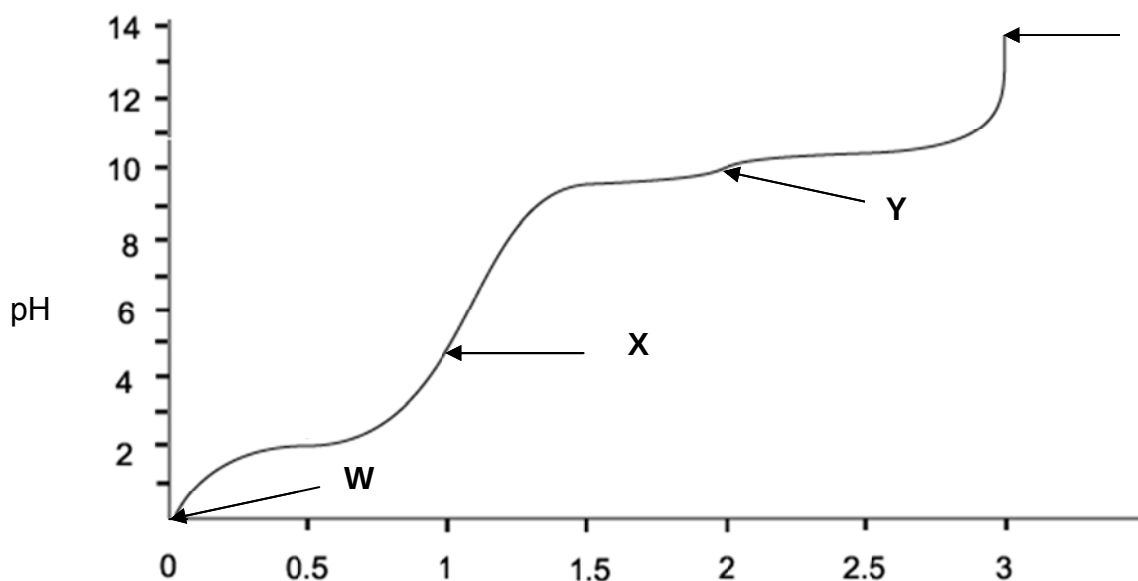
[6]

- (b)** A buffer solution is prepared by adding 10.00 cm³ of 0.100 mol dm⁻³ potassium hydroxide to 25.00 cm³ of 0.400 mol dm⁻³ ethanoic acid. (pK_a of ethanoic acid is 4.75)

- (i) What is meant by the term *buffer*?
- (ii) Explain how the solution above acts as a buffer when H⁺ ions are added.
- (iii) Calculate the pH of the buffer solution formed.
- (iv) Suggest why the concentration of potassium hydroxide in a solution slowly decreases when left open to air.

[6]

- (c) A solution was prepared by reacting 1 mol of lysine with 2 mol of hydrochloric acid. This solution was titrated with aqueous sodium hydroxide. The titration curve obtained is shown below.



- (i) Draw the structure of the species present at point W, X, Y and Z on the curve.
- (ii) Define *isoelectric point*.
- (iii) Use the titration curve to suggest the *isoelectric point*, and the pK_a values of the alpha-carboxyl group, the alpha-amino group and the amino group of the side chain of lysine.
- (iv) State the number of different compounds (exclude stereoisomers) that could be formed when one molecule of lysine reacts with one molecule of proline.

line :

[8]
[Total: 20]

[Turn over

3(a) At standard temperature and pressure, oxygen is an odourless gas with the molecular formula O_2 . Oxygen may also be found in the form of oxide, O^{2-} or ozone, O_3 .

(i) Sketch a graph of the pH of the solution when the highest oxide of the elements from sodium to sulfur are added separately to water.

(ii) With the aid of equations, state and explain the resulting pH of the solution for these pair of compounds when they are added separately to water.

1 SiO_2 and $SiCl_4$

2 Al_2O_3 and $AlCl_3$

(iii) Explain the following observations as fully as you can.

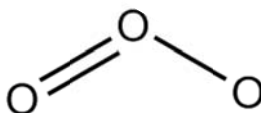
1 The first electron affinity of oxygen is exothermic while the second electron affinity of oxygen is endothermic.

2 The first ionisation of oxygen is less endothermic than that of the preceding element in the Periodic Table.

3 The boiling point of oxygen is lower than that of ozone.

[10]

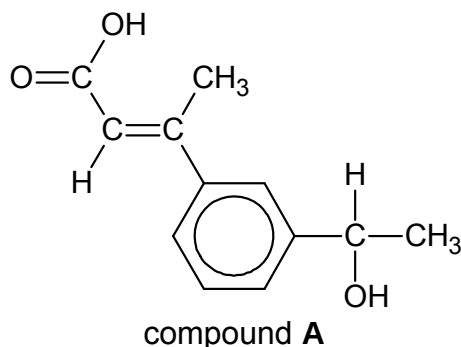
(b) The diagram below shows the structure of O_3 .



(i) The O-O bond length is 148 pm while the O=O bond length is 121 pm.

Suggest a reason why the bond length between the oxygen atoms in ozone is actually 128 pm. ($1 \text{ pm} = 10^{-12} \text{ m}$)

Both manganate(VII) ion and ozone can cause oxidative cleavage of alkenes.



- (ii) Draw the structural formulae of the products formed when compound **A** reacts with hot acidified aqueous potassium manganate(VII).
- (iii) When ozone reacts with compound **A**, one of the products gives a positive result with ammoniacal silver nitrate solution. Suggest the structural formulae of the products formed.

When ozone reacts with nitrogen dioxide, gaseous N_2O_5 and oxygen gas is produced. N_2O_5 is a rare example of a compound that adopts two structures depending on the conditions: as an ionic salt when in the solid state, but a molecule when in the gaseous state.

- (iv) Gaseous N_2O_5 is a symmetrical molecule with the two nitrogen atoms bonded to one central oxygen atom.

Draw a dot-and-cross diagram to show the bonding in N_2O_5 molecule. Use your diagram to suggest the shape and the bond angle with respect to nitrogen. Determine whether the molecule is polar or non-polar.

- (v) Solid N_2O_5 contains a cation that can react with benzene via a substitution reaction.

Suggest the identity of the cation and anion present in the solid N_2O_5 .

- (vi) N_2O_5 decomposes completely to give nitrogen dioxide and oxygen.

Write an equation for this reaction.

[10]
[Total: 20]

[Turn over

4 Nitric oxide is a chemical compound with chemical formula NO. It is an extremely important intermediate in the chemical industry. Nitric oxide is an air pollutant and it reacts with water to form nitrous acid, HNO₂, and nitric acid, HNO₃.

(a) In the laboratory, nitric oxide is conveniently generated by reduction of nitric acid with copper and has the overall $E^\ominus_{\text{cell}}$ value of + 0.62 V.

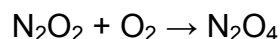
- (i)** Write the half-equation for the reduction of nitric acid to nitric oxide and determine its standard electrode potential.
- (ii)** Explain the effect of using copper powder of the same mass on the rate and feasibility of the reaction.

[4]

(b) The non-elementary reaction between NO and O₂ is a third order reaction and N₂O₄ is formed as the product. The first step involves the formation of the unstable dimer, N₂O₂.



This is followed by the rate-determining step.



- (i)** Draw a diagram to show the shape and bonding in N₂O₂ and hence suggest a reason why N₂O₂ is unstable.
- (ii)** Using data from *Data Booklet* and relevant information given, calculate the bond energy of nitrogen-oxygen bond in NO.

Bond energy of nitrogen-oxygen bond in N₂O₂ is 607 kJ mol⁻¹.

- (iii)** Determine the rate equation for the reaction between NO and O₂.
- (iv)** Suggest why is there an unusual observation that the reaction between NO and O₂ becomes slower as the temperature is increased?

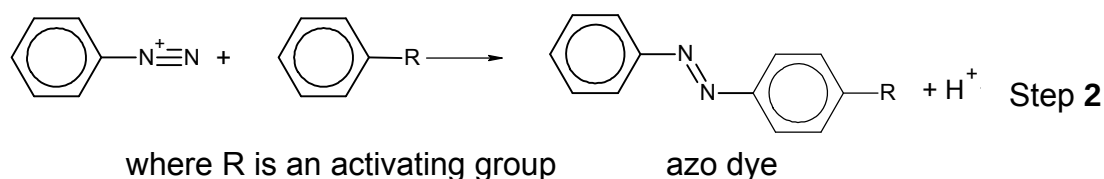
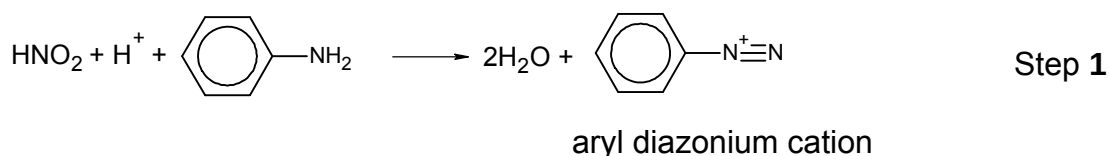
[6]

(c) Nitric oxide reacts with potassium hydroxide to produce potassium nitrite, KNO₂, dinitrogen oxide, N₂O, and water.

- (i)** Write a balanced equation for this reaction.
- (ii)** State the type of reaction in **(c)(i)**.

[2]

- (d) Nitrous acid is involved in the synthesis of azo dyes such as methyl orange. The first step involves the reaction of nitrous acid with phenylamine to form aryl diazonium cation. The second step involves the reaction between the cation and another arene compound to form the azo dye.



- (i) Name the type of the reaction for step 2.
- (ii) It is possible to form another isomer of the azo dye in step 2.

Draw the structure of this isomer and suggest why this isomer is less likely to be formed.

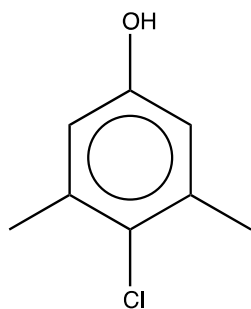
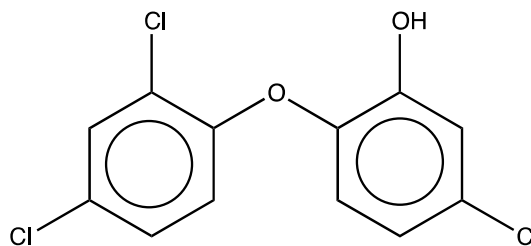
When nitrous acid reacts with propylamine, an alcohol and a neutral diatomic gas is obtained instead of the corresponding alkyl diazonium cation.

- (iii) Write a balanced equation for this reaction.
- (iv) Suggest why the alkyl diazonium cation is not obtained.
- (v) Optically active compound **A**, $\text{C}_3\text{H}_8\text{NOCl}$, contains three different functional groups. One of the functional groups is amine. **A** reacts with nitrous acid to form compound **B**, $\text{C}_3\text{H}_7\text{O}_2\text{Cl}$ which is optically inactive.

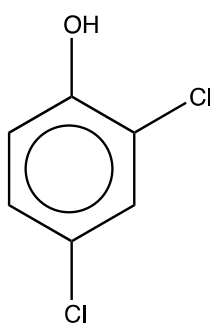
Draw the structures of **A** and **B**.

[8]
[Total: 20]

- 5 The compound *chloroxylenol* and *Triclosan* are two chlorinated phenol compounds which has been featured for its anti-microbial action, and used in soaps and detergents. While these compounds are relatively safe for humans to handle, they may have a tendency to persist in the environment for a long period of time, where in turn, they can be converted to other more toxic compounds, causing environmental problems.

*chloroxylenol**Triclosan*

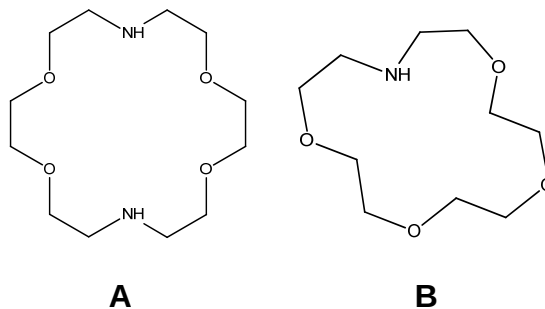
- (a) By considering their solubility and reaction with water, suggest two reasons why such compounds are likely to persist in the environment. [2]
- (b) (i) Suggest suitable reagents and conditions which would convert 3,5-dimethylphenol to *chloroxylenol*.
 (ii) State and outline the mechanism by which the reaction in (b)(i) is likely to occur. [3]
- (c) *Triclosan* can be synthesised from 2,4-dichlorophenol.



Compare the acidity of 2,4-dichlorophenol with *chloroxylenol*.

[2]

A compound related to 2,4-dichlorophenol and *chloroxyleneol* known as catechol, can be found in small quantities in fruits and vegetables. When mixed with an enzyme and with exposure to oxygen, catechol is converted to brown melanoid pigments, which accounts for why these fruits would turn brown when cut and left exposed to air. In the laboratory, this oxidation can be studied through the use of suitable metal complexes (e.g. Zn^{2+} , Ni^{2+}) in the presence of azacrown ethers, such as **A** and **B** shown below.



- (d) Using your knowledge of the rate of enzymatic activity, sketch a graph to illustrate how the rate of 'browning' of fruits change with increasing concentration of catechol.

[1]

- (e) (i) Draw the structure of complex formed from the reaction between compound **A** with Ni^{2+} respectively, indicating clearly its shape.
- (ii) Suggest the likely shape of the complex formed between Ni^{2+} and compound **B**.

[2]

Compounds **C** and **D** are isomers of catechol, and they have a formula of $\text{C}_6\text{H}_6\text{O}_2$. These compounds are insoluble in water and sodium carbonate, but soluble in sodium hydroxide.

Upon careful warming with potassium manganate(VII), catechol and compounds **C** and **D** produce 3 compounds labelled **E**, **F** and **G** respectively. Compounds **E**, **F** and **G** all give an orange precipitate upon warming with 2,4-dinitrophenylhydrazine. In addition, 1 mole of each compound decolourises 2 moles of bromine dissolved in tetrachloromethane.

However combustion of molar quantities of **G** requires half a molar equivalent of oxygen gas lesser compared to combustion of **E** and **F**. Also upon further vigorous heating which breaks down their ring structure, compounds of **F** and **G** give rise to two molecules of $\text{C}_3\text{O}_5\text{H}_2$, whereas **E** produces $\text{C}_4\text{O}_6\text{H}_2$.

- (f) Suggest the structures of catechol and compounds **C** to **G**, explaining clearly the chemistry involved and your reasoning.

[10]

[Total: 20]

END OF PAPER

[Turn over]

& 2

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Answer any 4 out of 5 questions in Paper 3.

Begin each question on a *fresh page* of writing paper.

Fasten your answers for **Question 1 and 2** behind this Cover Page.

You are advised to spend about **30 min per question**.

Circle the questions that you attempted.

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FOR EXAMINER'S USE		
Paper 3	Q3	/ 20
	Q4	/ 20
	Q5	/ 20